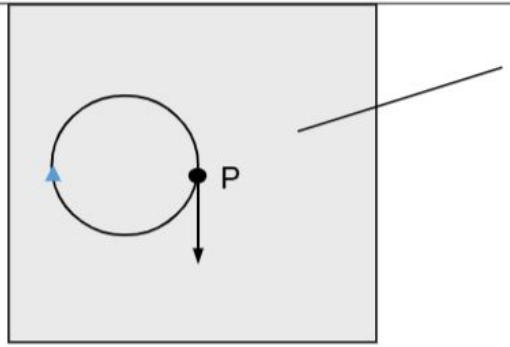
	successive decrease in peak	B1
5ai	Change in moment due to W = $mg \times (\Delta)L$ $= 1.3 \times 10^{-4} \times 9.81 \times 2.6 \times 10^{-2}$ $= 3.32 \times 10^{-5}$ $= 3.3 \times 10^{-5}$ N m	A1

5aii	Change in moment due to W = Change in moment due to magnetic force on QL $3.3 \times 10^{-5} = B \times (\Delta)I \times L \times x$ $3.3 \times 10^{-5} = B \times 1.2 \times 0.85 \times 10^{-2} \times 5.6 \times 10^{-2}$ B = 0.058 T	 C1 A1
5bi	 Circular path with diameter of circle perpendicular to velocity at P. Direction of path is clockwise.	 B1 B1
5bii	magnetic force provides the centripetal force $Bev = mv^2 / r$ $r = mv / Be$ $= (9.11 \times 10^{-31})(2.9 \times 10^{7}) / (1.5 \times 10^{-3}) (1.60 \times 10^{-19})$ = 0.110 m	 M1 A1
5biii	(With B and E field into the plane of the paper, the path of the electron will be) a <u>helix</u> <u>with an increasing pitch/vertical separation</u> and moving <u>out of the plane of the paper</u>.	 B1 B1
6a	The <u>changing/alternating</u> current in <u>coil C</u> produces a <u>changing/alternating magnetic</u> flux density (*B*) (since magnetic flux density is directly proportional to the current)	B1